TASK 2: METHODOLOGY PASSPORT CARD

Research Methods

Yerkindik Ailen | Aitkali Dariga

1. Project Information

Research title: The Impact of Generative AI Assistance on Programming Learning Effectiveness Among Undergraduate Students

GitHub repository: https://github.com/ailerkk/AI-Programming-Research

Research aim: To examine how generative AI affects programming learning among undergraduate students and identify which type of AI support helps students learn without reducing their ability to solve problems independently.

2. Research Type and Study Design

Research type: Quantitative research

We plan to compare students in three conditions: no AI assistance, AI hints and explanations, and direct AI-generated solutions. We will use the same programming language, similar tasks, time limits, and scoring rules for all groups. Prior programming experience will be recorded so that we can consider differences between students.

During practice, students will work under their assigned condition. After that, all students will complete a separate programming task without AI. This will help us compare both performance during practice and independent performance afterward. A later concept test will be used to assess retention if the study is carried out.

3. Research Questions and Hypotheses

Research Question 1

How does access to generative AI during programming exercises affect students' task performance and completion time compared with learning without AI?

H0: There is no statistically significant difference in programming task scores or completion time between students using generative AI and students working without AI.

H1: Students using generative AI achieve higher programming task scores and complete tasks faster than students working without AI.

Research Question 2

How does the use of generative AI during programming practice affect students' understanding and retention when AI assistance is later removed?

H0: There is no statistically significant difference in understanding and retention between students who practiced with AI and students who practiced without AI.

H1: Students who practiced with AI show a statistically significant difference in understanding and retention when completing later tasks without AI.

Research Question 3

Which type of AI support, explanations and hints or direct solution generation, leads to better independent programming performance?

H0: There is no statistically significant difference in independent programming performance between students receiving AI-generated hints and students receiving complete AI-generated solutions.

H1: Students receiving AI-generated hints and explanations achieve higher independent programming performance than students receiving complete AI-generated solutions.

4. Variables Matrix

Variable type	Variables	How we will use them
Independent	AI access; type of AI support	Three groups: no AI, AI hints, direct AI solutions
Dependent	Task score (%); completion time (minutes); independent task score (%); programming errors (count); concept test score (%)	Scores and time recorded for each student
Controlled	Programming language; task difficulty; time limit; learning materials; evaluation criteria	Kept as similar as possible for all groups
Background factor	Prior programming experience	Measured before the experiment and considered in analysis

5. Metrics and Baseline

5.1. Primary Metric

Independent Programming Task Score (%): The percentage of test cases passed on a task completed without AI assistance.

Formula: Independent Task Score (%) = (Passed Test Cases / Total Test Cases) × 100

We selected this as the main metric because the research focuses on whether students can solve programming problems on their own.

5.2. Secondary and Guardrail Metrics

Metric	Definition / unit	Purpose
Completion time	Minutes needed to finish the task	Compare task speed
Concept test score	Correct answers / total answers × 100 (%)	Check understanding
Programming errors	Number of recorded errors	Compare mistakes
Execution error rate (guardrail)	Programs that fail to execute / submitted programs × 100 (%)	Monitor code execution problems
Valid results rate (guardrail)	Valid records / all records × 100 (%)	Monitor data quality
Pipeline reliability (guardrail)	Benchmark finishes without a critical error	Confirm the script can run

The current demonstration script calculates task scores, completion time, concept scores, and programming errors. The other guardrail measures are planned for the full data collection stage and are not yet calculated automatically.

5.3. Baseline

Group A — No AI: Students complete programming exercises without generative AI. This is our baseline.

Group B — AI hints: Students can use AI-generated explanations and hints, but not complete solutions.

Group C — AI solutions: Students can use direct AI-generated solutions during practice.

After the practice phase, all groups will complete an independent task without AI. We will compare their results with the baseline and with each other.

6. Reproducibility and Current Project Status

We created a public GitHub repository with a README, configuration file, sample CSV data, Python benchmark, pinned dependency version, Dockerfile, and an automated GitHub Actions workflow.

The sample dataset contains nine synthetic records (three per group). These are made-up examples used only to check whether the program works. They are not real student results and cannot be used to accept or reject our hypotheses.

The benchmark successfully ran in GitHub Actions. It calculated group averages from the sample file, and the automated workflow also completed the Docker build and run steps successfully.

Group (synthetic data)	Mean task score	Mean time	Mean concept score	Mean errors
No AI	70.00%	35.67 min	70.00%	2.00
AI hints	86.67%	25.67 min	86.67%	0.67
AI solutions	76.67%	20.33 min	66.67%	1.33

These numbers are demonstration outputs only. Real data collection, hypothesis testing, and conclusions will come later.

7. Files and Reproduction

Main files: README.md; src/benchmark.py; configs/config.yaml; data/sample/sample_results.csv; requirements.txt; Dockerfile; .github/workflows/test.yml; .gitignore; LICENSE.

To run the project with Docker from the repository root:

docker build -t ai-programming-research .

docker run --rm ai-programming-research

The script reads the synthetic CSV file, calculates average metrics for the three groups, and prints the results. The same check is run automatically through GitHub Actions.